

35.0.1 NIELIT 2022 April Scientist B | Section B | Question: 117



Consider sinusoidal modulation in an AM system. Assuming no over modulation, the modulation index (μ) when the maximum and minimum values of the envelope, respectively, are 3 V and 1 V is _____.

- A. 0.1 B. 0 C. 0.3 D. 0.5

nielit2022apr-scientistb analytical-aptitude

35.1

Cache Memory (1)

35.1.1 Cache Memory: NIELIT 2021 Dec Scientist B - Section B: 22



Cache memory works on the principle of:

- A. Locality of data B. Locality of memory
C. Locality of reference D. Locality of reference & memory

nielit-2021-it-dec-scientistb cache-memory co-and-architecture

Answer key

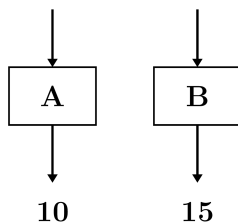
35.2

Cyclomatic Complexity (2)

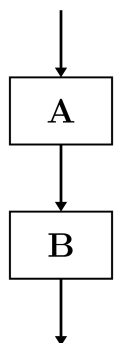
35.2.1 Cyclomatic Complexity: NIELIT 2021 Dec Scientist A - Section B: 63



The Cyclomatic complexity of two modules A and B are 10 and 15 respectively :



What is the cyclomatic complexity of sequential integration of A and B ?



- A. 19 B. 21 C. 24 D. 25

nielit2021dec-scientista is&software-engineering cyclomatic-complexity logical-reasoning

35.2.2 Cyclomatic Complexity: NIELIT 2021 Dec Scientist B - Section B: 48



Which testing is equivalent to Cyclomatic complexity?

- A. Black box testing
- B. Green box testing
- C. Yellow box testing
- D. White box testing

nielit-2021-it-dec-scientistb software-testing cyclomatic-complexity

Answer key

35.3 Effective Memory Access (2)

35.3.1 Effective Memory Access: NIELIT 2021 Dec Scientist B - Section B: 113



When a cache is 10 times faster than main memory, and the cache can be used 70% of the time, how much speed can be gained using cache ?

- A. $\simeq 10$
- B. $\simeq 0.3$
- C. $\simeq 0.7$
- D. $\simeq 2.7$

nielit2021dec-scientistb cache-memory effective-memory-access co-and-architecture

35.3.2 Effective Memory Access: NIELIT 2022 April Scientist B | Section B | Question: 102



Consider a machine with 40 MHz processor which has run a benchmark program. The executed program consists of 100,000 instruction executions, with the following instructions mix and clock cycle count. What will be the effective CPI, MIPS rate, and execution time.

Instruction Type	Instruction Count	Cycles/Instructions
Integer arithmetic	45000	1
Data Transfer	32000	2
Floating point	15000	2
Control transfer	8000	2

- A. CPI : 3.55; MIPS : 30; Execution time : 1.87 ms
- B. CPI : 1.55; MIPS : 25.8; Execution time : 3.87 ms
- C. CPI : 5.60; MIPS : 45.8; Execution time : 2.87 ms
- D. CPI : 2.55; MIPS : 35.8; Execution time : 4.87 ms

nielit2022apr-scientistb co-and-architecture effective-memory-access

Answer key

35.4 Group Theory (1)

35.4.1 Group Theory: NIELIT 2022 April Scientist B | Section B | Question: 65



Let G be a multiplicative group and $a \in G$. If the order of a is 6, then the order of a^5 is equal to :

- A. 1 B. 5 C. 6 D. 30

nielit2022apr-scientistb group-theory functions

Answer key

35.5

Is&software Engineering (1)

35.5.1 Is&software Engineering: NIELIT 2021 Dec Scientist B - Section B: 24



Consider a program with following data:

Unique operator = 10, unique operands = 15

Total operator = 30, Total operands = 40

What is the estimated length of program?

- A. 132 B. 92 C. 32 D. 82

nielit-2021-it-dec-scientistb is&software-engineering

Answer key

35.6

Percentage (1)

35.6.1 Percentage: NIELIT 2021 Dec Scientist A - Section B: 62



What is the annual change in traffic of software with 1 million lines of code with 30% lines added and 10% lines are deleted ?

- A. 0.25 B. 0.15 C. 0.4 D. 0.6

nielit2021dec-scientista is&software-engineering percentage quantitative-aptitude

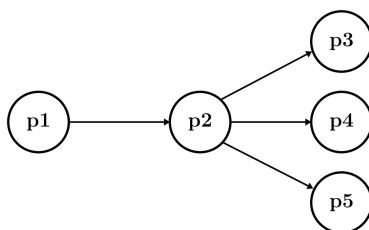
35.7

Software Design (1)

35.7.1 Software Design: NIELIT 2022 April Scientist B | Section B | Question: 79



In a data flow diagram, the segment shown below is identified as having transaction flow characteristics, with $p2$ identified as the transaction center



A first level architectural design of this segment will result in a set of process modules with an associated invocation sequence. The most appropriate architecture is

- A. p_1 invokes p_2 , p_2 invokes either p_3 , or p_4 , or p_5
- B. p_2 invokes p_1 and then invokes p_3 , or p_4 , or p_5
- C. A new module T_c is defined to control the transaction flow. This module T_c first invokes p_1 and then invokes p_2 . p_2 then invokes p_3 , or p_4 , or p_5
- D. A new module T_c is defined to control the transaction flow. This module T_c invokes p_2 . p_2 invokes p_1 , and then invokes p_3 , or p_4 , or p_5

nielit2022apr-scientistb is&software-engineering software-design

35.8

Software Testing (1)

35.8.1 Software Testing: NIELIT 2021 Dec Scientist B - Section B: 47



Maintenance testing is performed using which methodology?

- A. Breadth test and depth test
- B. Sanity testing
- C. Retesting
- D. Confirmation testing

nielit-2021-it-dec-scientistb is&software-engineering software-testing

Answer key

Answer Keys

35.0.1	D	35.1.1	C	35.2.1	C	35.2.2	D	35.3.1	D
35.3.2	B	35.4.1	C	35.5.1	B	35.6.1	C	35.7.1	C
35.8.1	A								